RTML MEETING

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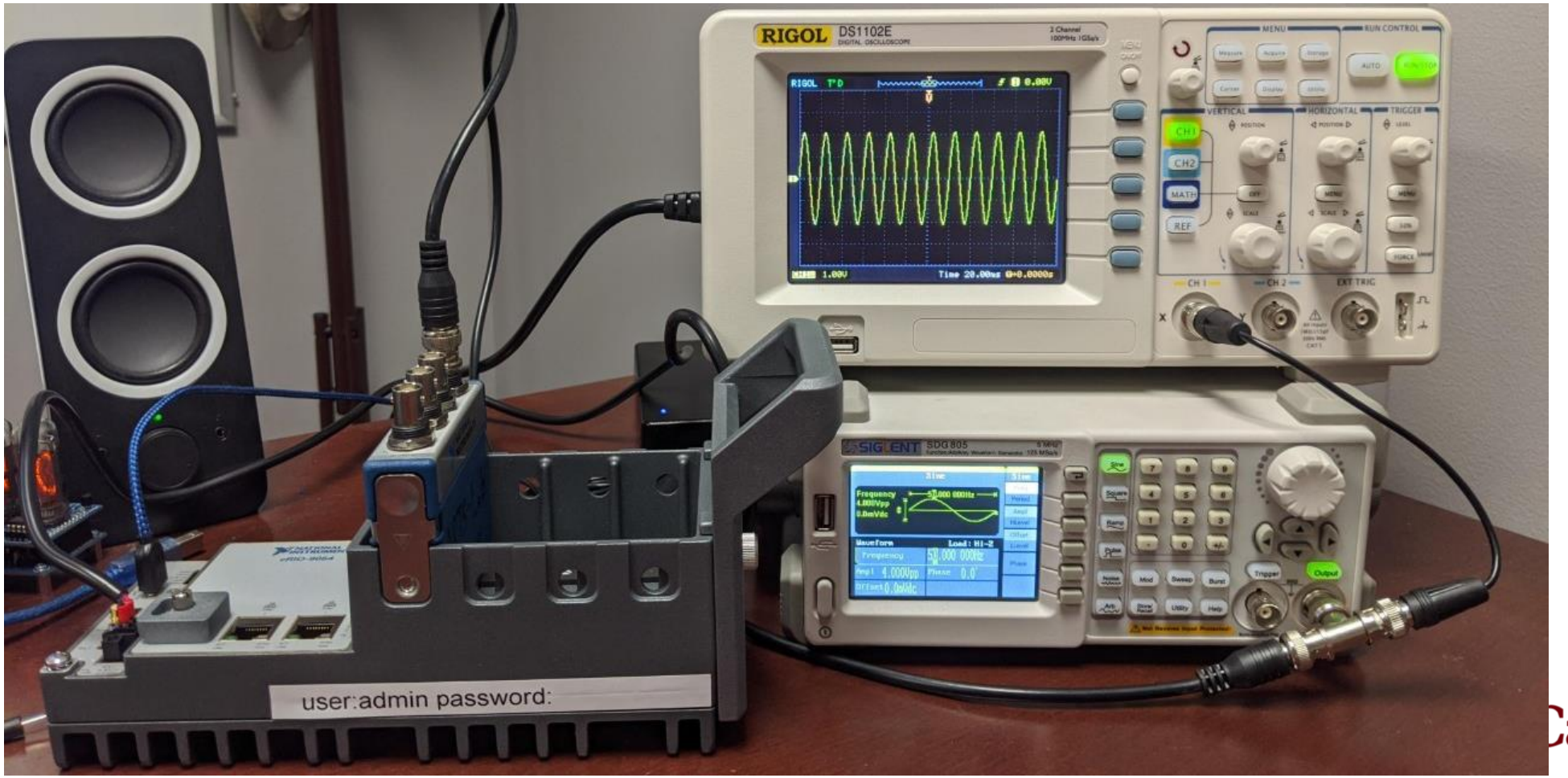
South Carolina

FFT running on FPGA

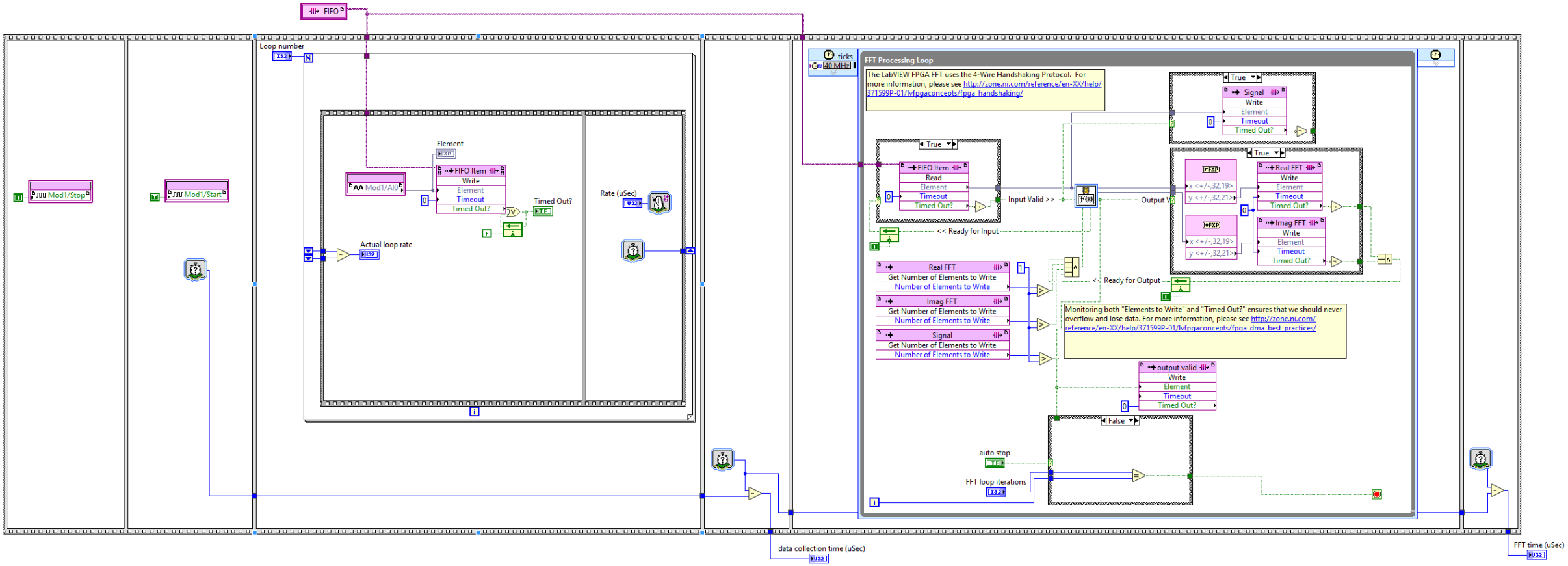
- 24-bit +/- 60v ADC with max rate of 50k S/s
- 1.33 GHz Dual-Core CPU, 4 GB Storage, Artix 7 100T FPGA



Carolina

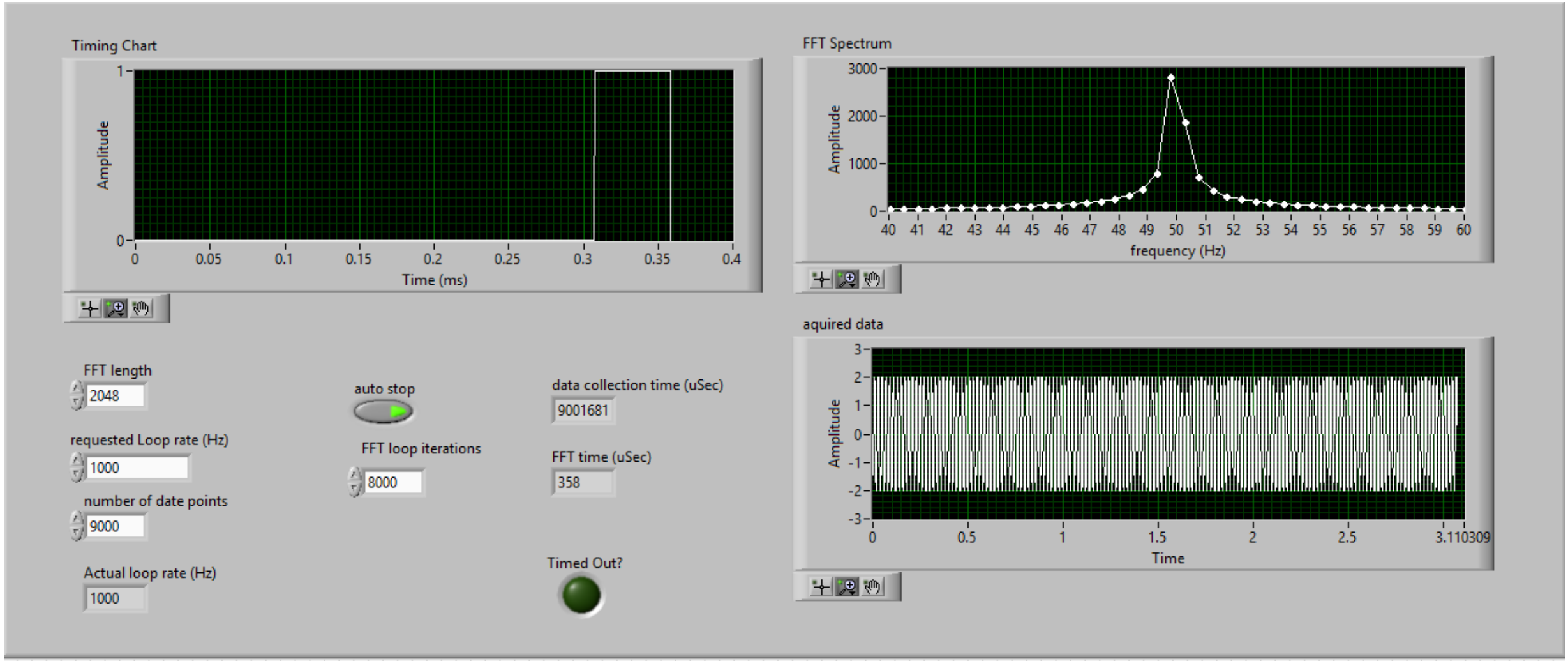
FFT running on FPGA

- 40 MHz clock
- 5-7 ticks / sample
- Could use 1 tick / sample if needed
- Limited to a FFT length of 8192 with current code, 65536 is max value otherwise.



FFT length 2048

- 0.05 ms compute time for FFT



FFT length 2048

- 0.05 ms compute time

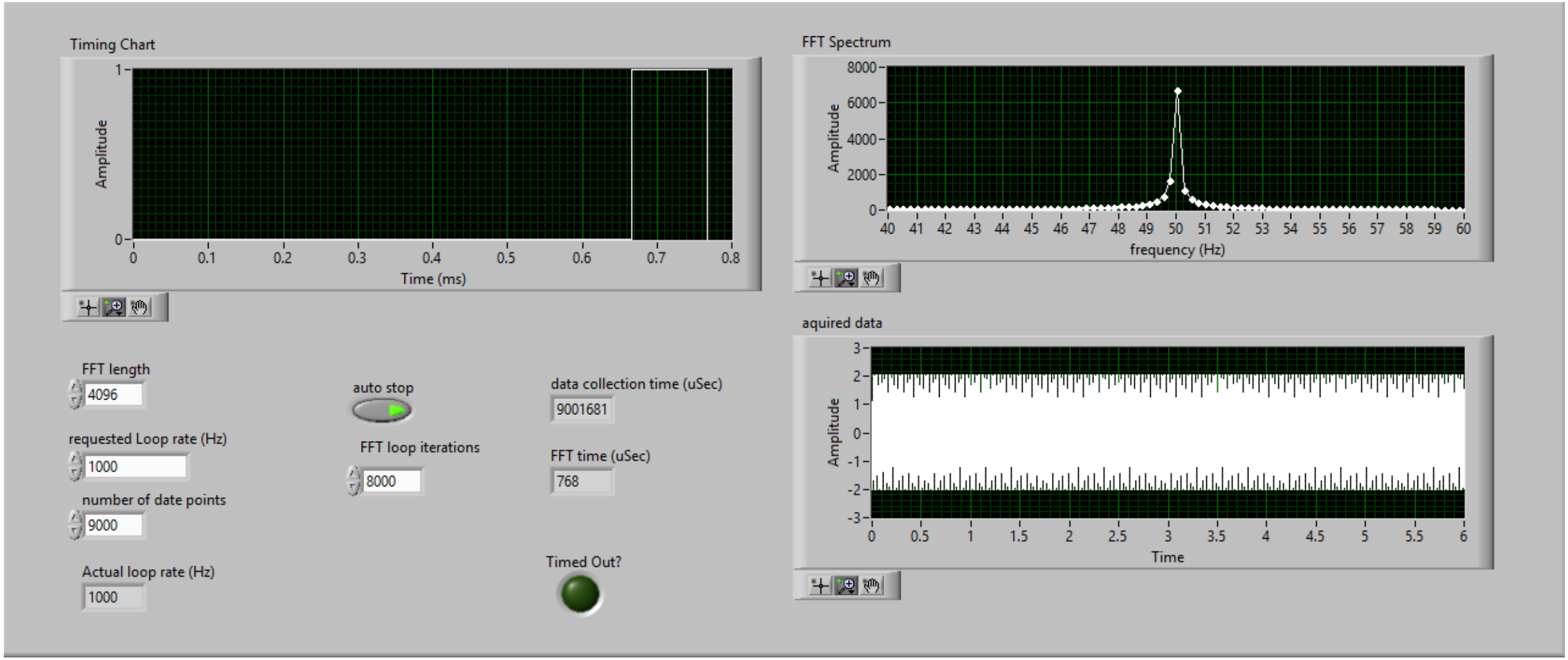
Reports

Final device utilization (placement)

Device Utilization	Used	Total	Percent
Total Slices	4583	15850	28.9
Slice Registers	12159	126800	9.6
Slice LUTs	11240	63400	17.7
Block RAMs	84	135	62.2
DSP48s	17	240	7.1

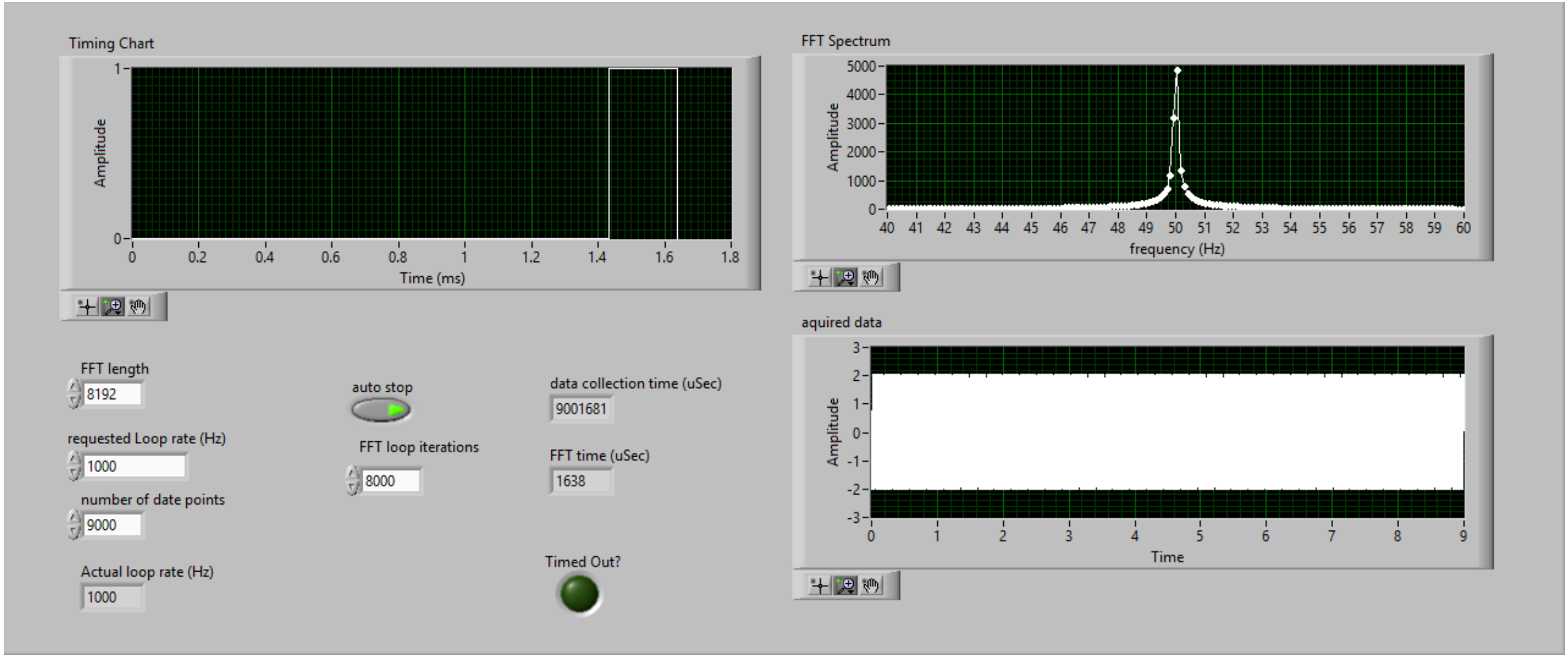
FFT length 4096

- 0.1 ms compute time for FFT



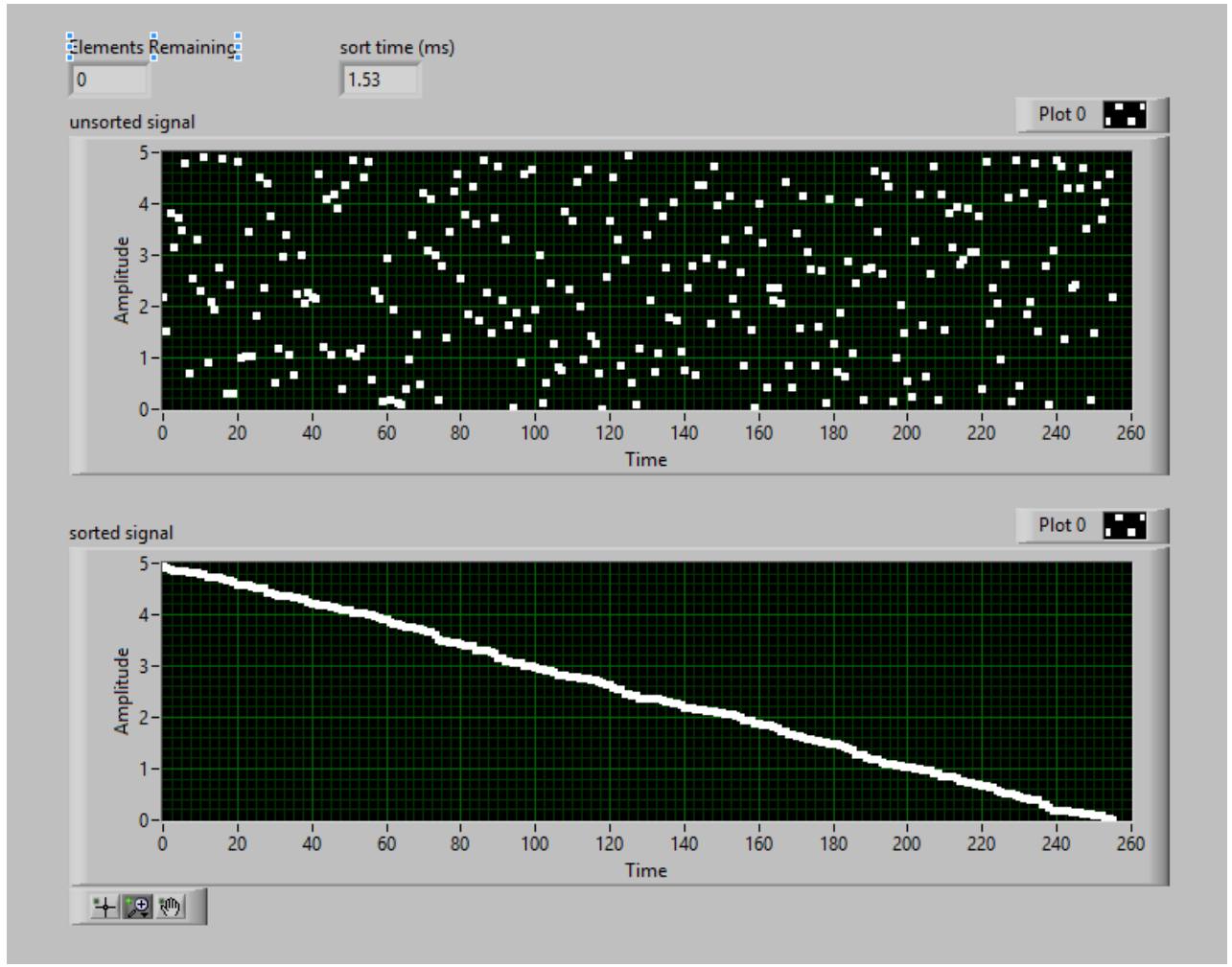
FFT length 8192

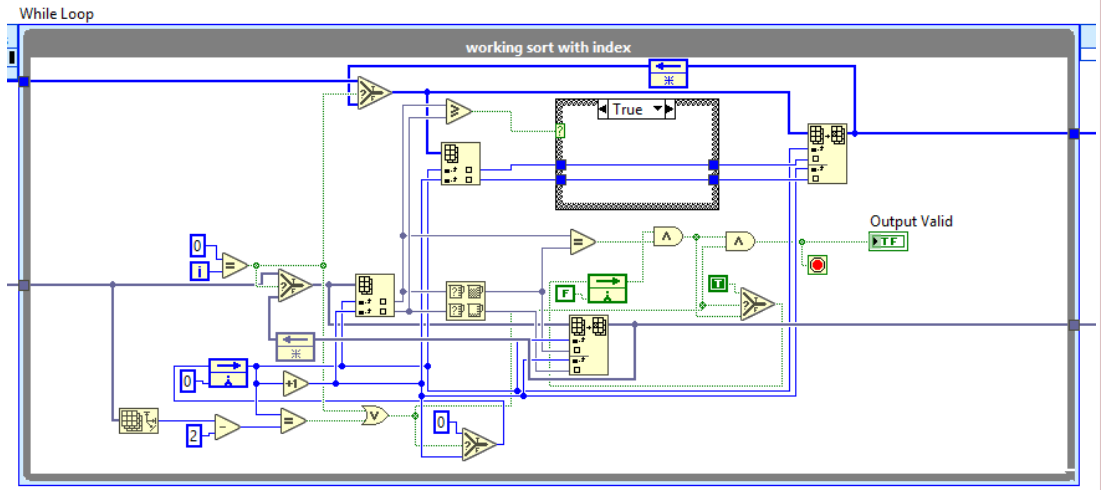
- 0.2 ms compute time for FFT



Bubble sort 254

- 1.53 ms compute time





Desktop sort 254

- 0.056 ms compute time
- np.sort() uses quicksort

South Carolina

Bubble sort 254 points

- 40 MHz clock
- Could not compile higher

Compilation Status

Build Specifications

FPGA_bubble_SCTL_FIFO (Sort_Pro

Status

Complete

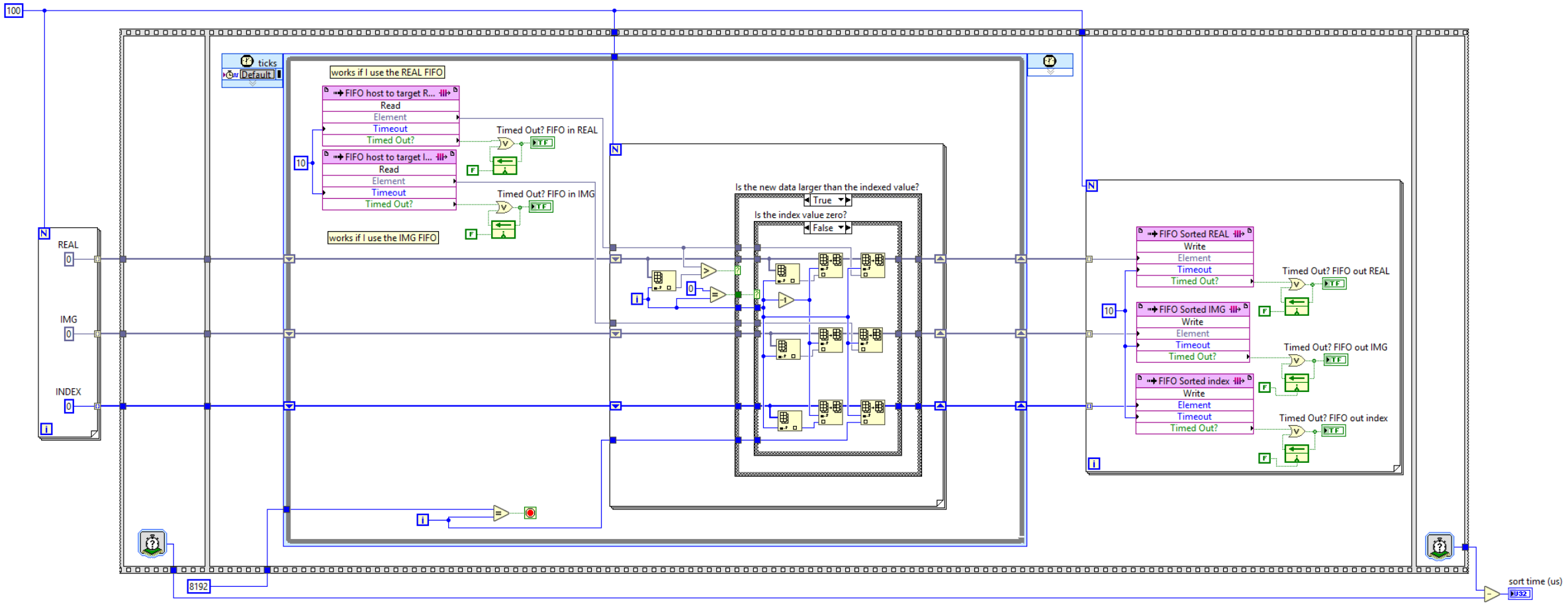
Reports

Final device utilization (placement)

Device Utilization	Used	Total	Percent
Total Slices	14639	15850	92.4
Slice Registers	49818	126800	39.3
Slice LUTs	42753	63400	67.4
Block RAMs	29	135	21.5
DSP48s	0	240	0.0

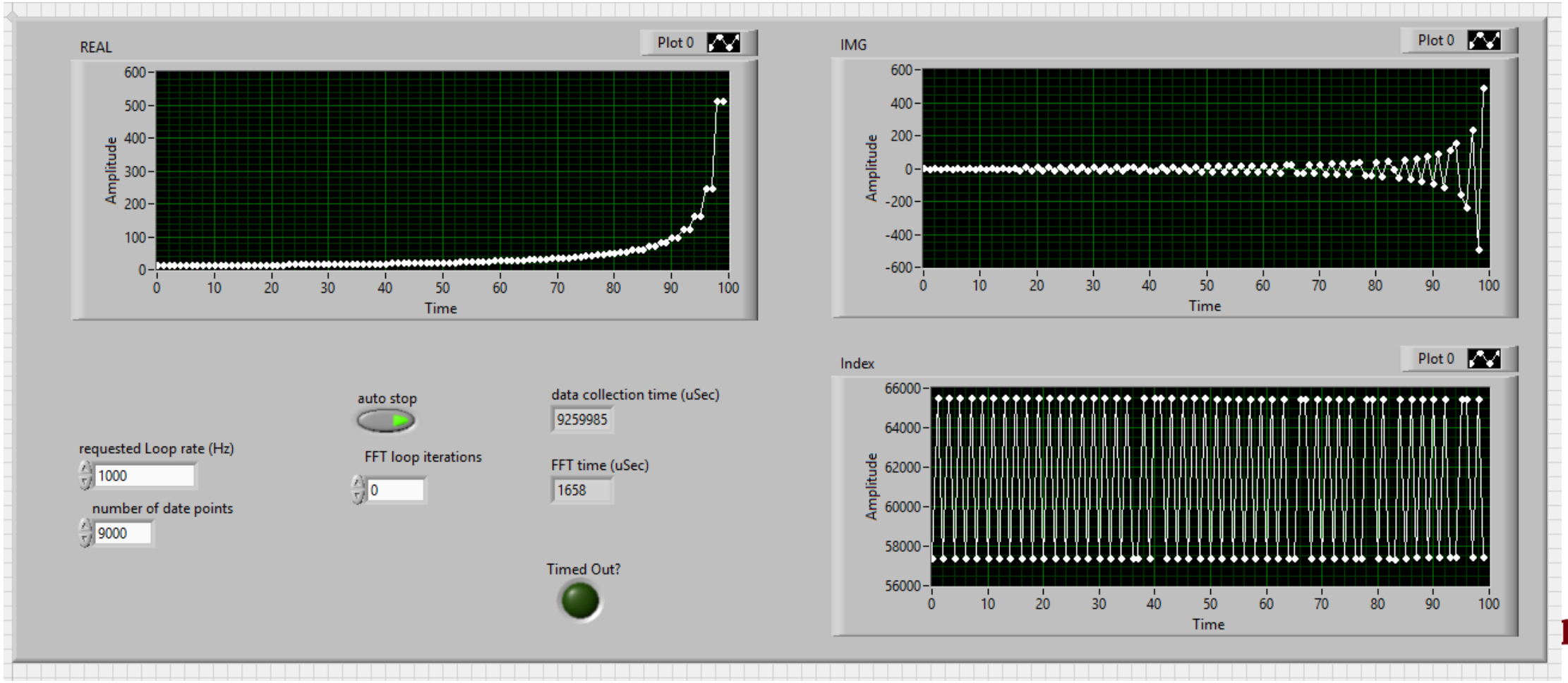
Catch the top *n* points

- Stores the top n points
- Includes real, Img, and fft index
- 200 us for 100 points



Catch the top *n* points

- 8192 FFT points, stores top 100 points
- 1.68 ms



Catch the top *n* points

- 40 MHz clock

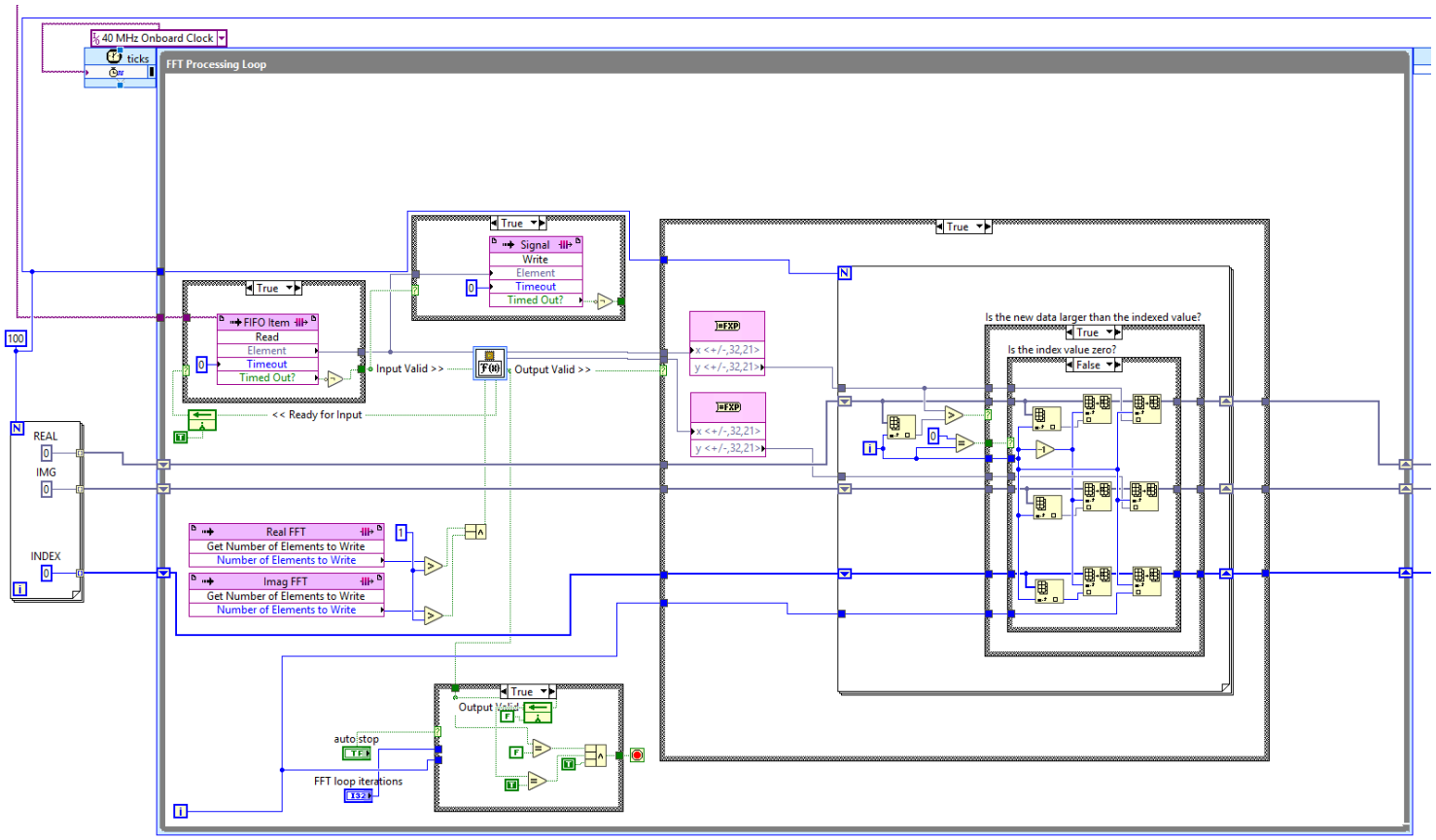
Status

Complete

Reports

Final device utilization (placement)

Device Utilization	Used	Total	Percent
Total Slices	12797	15850	80.7
Slice Registers	31231	126800	24.6
Slice LUTs	43047	63400	67.9
Block RAMs	98	135	72.6
DSP48s	17	240	7.1



Bubble sort 254 points

- 40 MHz clock
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THANKS!